

Vahe Gdlyan

ML Engineer & Data Scientist · Production Systems · Computer Vision · Statistical Experimentation
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ENGINEERING PROJECTS

NucleAI — Interactive Energy Simulator & Verifiable AI

May 2026

Lead Engineer | Python · FastAPI · Streamlit · ChromaDB · OpenAI API · JavaScript

- Engineered a bifurcated architecture: a **FastAPI RAG pipeline** for IAEA-grounded AI Q&A, and a parallel **vanilla JavaScript engine** executing piecewise-linear mathematical models for dynamic grid stability, lifecycle CO₂ emissions, and cost penalties.
- Authored a custom pure-Python chunking and ingestion script (chunk_size=1000, overlap=200, paragraph → line → word cascade) to embed **10 IAEA PDFs** into a local ChromaDB vector store; metadata-tagged per source filename and page number for citation traceability.
- Implemented top-k cosine similarity retrieval (k=4) over **text-embedding-3-small** vectors to enforce hard context grounding for gpt-4o-mini completions; zero hallucination on in-distribution nuclear safety queries.
- Deployed **Streamlit multi-page frontend** (Altair charts, glassmorphism UI) consuming FastAPI endpoints for radiation dose comparison, Armenia energy mix visualization, and CO₂ lifecycle analysis.
- JavaScript grid engine executes **8-branch narrative priority cascade** and a 4-component 100-point scoring system (output, emissions, stability, diversification) with grade mapping; zero external dependencies.

Vestimate — Multi-Cloud AI Wardrobe Intelligence Platform

Jan. 2025 – Present

Lead Architect | FastAPI · Flutter · Redis · Celery · Modal.com · Supabase · FashionCLIP · U-2-Net

- Architected full distributed ML system: FastAPI async gateway (202 Accepted), Redis/Celery task queue, Modal.com serverless NVIDIA T4 GPU inference, Supabase PostgreSQL + pgvector, Cloudflare R2 — operational at ≤\$80/month via scale-to-zero design.
- Dual-model GPU inference pipeline containerising **U-2-Net** (nested encoder-decoder salient object detection) + **FashionCLIP** (ViT-B/16, 2.5M+ fashion pairs) in single Modal container; warm latency ~1.2s/garment.
- Engineered **512-dim semantic vector store** via pgvector IVFFlat cosine indexing (100 lists); similarity queries <15ms across 1,000+ item wardrobes.
- Dual-stage RAG engine: Stage 1 pure pgvector retrieval; Stage 2 gpt-4o-mini synthesis with Python hallucination guard cross-validating every UUID before delivery — **95% semantic consistency rate**.
- Zero-shot FashionCLIP classifier over natural language anchor dictionary (category, material, fit, 32 chromatic anchors); **85% pass rate at 0.70 confidence**; auto-routes failures to manual_review_queue.
- Redis cache layer (Index 1) achieving <80ms on repeat daily opens (98%+ hit rate), bypassing LLM inference entirely. Graceful degradation fallback: pure pgvector color-harmony + category matching when OpenAI/Google unreachable — Flutter client never crashes.
- Flutter client: Clean Architecture, Riverpod 3.0 reactive state, Hive offline persistence, native camera garment digitisation.

AgriVision — End-to-End CV Decision Support System

Jan. 2026 – Present

| YOLOv8 · PyTorch

- Focal boundary loss with adaptive photometric weighting**: eliminated 1,092 zero-IoU hallucinated detections, preventing 4–8% yield overestimation; maintained 0.944 mAP@50 on Global Wheat holdout while increasing median prediction confidence +39.4% and reducing training epochs by 30%.

- Authored 7-page mathematical diagnostic paper on Out-of-Distribution boundary analysis (preparing arXiv preprint). Deployed inference state engine via Hugging Face cloud server for spatial uniformity computation.

DATA SCIENCE & ANALYTICS

A/B Test Stress-Test Pipeline

Mar. – Apr. 2025

| Python · SciPy · Pandas

- 5-gate statistical validation pipeline across **887K users**; prevented deployment of false-positive +4.29% conversion lift ($p=0.1669$) by requiring significance survival across independent tests.
- Named the **"False Flat" failure mode**: +4.29% relative lift with 95% CI crossing zero, indistinguishable from noise under two-proportion Z-test. Applied Welch's T-test on zero-inflated RPV distribution (98%+ \$0 spenders) — surfaced statistically significant **+8.17% RPV lift** hidden by the primary metric.
- Simpson's Paradox pressure-test via weekday/weekend segmentation (63.5%/36.5%); confirmed structural stability with 0.21pp lift delta. Gini coefficient analysis confirmed revenue concentration drop $0.307 \rightarrow 0.291$ — lift distributed across user base, not whale-driven.

Zero-Inflated Revenue Prediction — Two-Stage Hurdle Model

- Architected Hurdle Model on **390K users** (1.69% payer rate): $P(\text{Payer}) \times E[\text{Revenue}|\text{Payer}]$; achieved **92.9% Revenue Capture Rate** in top predicted decile.
- Stage 1: LightGBM classifier (scale_pos_weight=58.1, GOSS, stratified 5-fold CV); diagnosed 4× probability over-confidence via **Platt Scaling**; reduced MAE \$8.79 $\rightarrow$ \$1.83 (79.2% improvement) with zero ranking degradation.
- Engineered **107 behavioral features** from 72-hour lifecycle window. Stage 2 XGBoost log-transform regression ($\log_{1p}$) on multi-modal distribution (\$14/\$53/\$79 tiers); threshold optimised across 86 candidate values to minimise production MAE.

Product Funnel & User Behavior Analysis

Feb. – Mar. 2025

- End-to-end funnel analysis across acquisition, activation, and monetisation stages; cohort-segmented by behavioral and demographic attributes to surface heterogeneous conversion patterns masked by aggregate metrics. Produced prioritised intervention diagnostics that seeded the A/B testing pipeline.

TECHNICAL WRITING

UnderTheHood Engineering — Medium Publication

Apr. 2025 – Present

- Publishes production-grade ML and engineering frameworks bridging academic statistical methodology and engineering decision-making. Writing is mathematical, first-principles, and tool-agnostic — focused on failure modes, calibration traps, and the gap between benchmark metrics and production behaviour.
- Covers experimentation pipelines, imbalanced classification, revenue modelling, and system design tradeoffs. Style prioritises rigour and reproducibility over narrative; each piece includes full mathematical derivations, concrete failure examples, and actionable engineering fixes.

EDUCATION & TRAINING

Picsart Academy — Machine Learning & Data Science Trainee

Feb. 2025 – Present

Production ML curriculum.

Abovyan Educational Complex — High School

Expected Graduation: 2026

Focus: Mathematics & Computer Science. Extracurricular: 2023 Vice Champion of Armenia, Junior Boxing.

AWARDS & RECOGNITION

 **Winner — Global HackAtom 2026, National Stage (Armenia)** May 2026

- Built and presented **NucleAI** — a live, bifurcated architecture combining a FastAPI RAG pipeline with a deterministic JavaScript grid simulation engine. Competed against national teams at the HackAtom Armenia 2026 selection stage.

1st Place — Plekhanov Russian University Startup & Innovation Competition May 2026

- Presented **Vestimate** as a live, operational multi-cloud AI system — sole entry running production infrastructure at time of judging. Competing teams submitted concepts and UI mockups; Vestimate demonstrated a fully containerised pipeline spanning six cloud platforms with measurable latency metrics (<80ms cache, <4s cold-path end-to-end, 95% LLM consistency) and a credible B2B roadmap. Unanimous 1st place across technical depth, commercial viability, and execution maturity.

TECHNICAL SKILLS

Languages: Python, SQL, Dart

ML Frameworks: PyTorch, LightGBM, XGBoost, Scikit-learn, HuggingFace Transformers, FashionCLIP, U-2-Net, YOLOv8

Backend & Infra: FastAPI, Uvicorn, Redis, Celery, Supabase (PostgreSQL + pgvector), ChromaDB, Cloudflare R2, Railway, Docker, REST API design, async task queues, scale-to-zero serverless GPU

Frontend & Mobile: Flutter (Clean Architecture, Riverpod 3.0, Hive offline persistence), Dart, Streamlit, JavaScript (vanilla)

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Altair, Plotly

Analytics & Experimentation: A/B testing, Z-test, Welch's T-test, Chi-square, SRM detection, Simpson's Paradox analysis, Gini coefficient, funnel analysis, cohort segmentation, causal inference, false positive prevention

ML Architectures: CNNs, RNNs, LSTMs, YOLO, Gradient Boosting, Vision Transformers (ViT), Contrastive Learning (CLIP), Dual-stage RAG, pgvector IVFFlat cosine indexing

Core Competencies: Custom loss function design, model calibration (Platt Scaling), revenue modelling, zero-inflated distribution handling, OOD analysis, distributed systems architecture, hallucination-resistant LLM pipeline engineering, vector database design